Chapter 5: Convex Optimisation

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Convex Optimisation Problems

A convex optimisation problem (or just a convex problem) is a problem consisting of minimizing a convex function $f(\mathbf{x})$ over a convex set C:

$$\begin{array}{ll} \min & f(\mathbf{x}) \\ \text{s.t.} & \mathbf{x} \in C \end{array} \tag{CVX}$$

A functional form of a convex problem can written as

$$\begin{array}{lll} \min & f(\mathbf{x}) & \\ \text{s.t.} & g_i(\mathbf{x}) \leq 0, & i = 1, 2, \ldots, m \\ & h_j(\mathbf{x}) = 0, & j = 1, 2, \ldots, p\,, \end{array}$$

where $f, g_1, \ldots, g_m : \mathbb{R}^n \longrightarrow \mathbb{R}$ are convex functions, and $h_1, h_2, \ldots, h_p : \mathbb{R}^m \longrightarrow \mathbb{R}$ are affine functions. The functional form does fit into the general formulation (CVX). A very important feature of convex optimization problems is that local minima are global minima!

Theorem (Local minima are global in CVX.)**.** *Let $f : C \longrightarrow \mathbb{R}$ be a convex function defined on the convex set $C \subseteq \mathbb{R}^n$. Let $\mathbf{x}^* \in C$ be a local minimum of f over C. Then x^* is a global minimum of f over C.*

Proof. Assume $\mathbf{x}^*$ is a local minimum of f over C. This implies that there exists $r > 0$ such that $f(\mathbf{x}) \geq f(\mathbf{x}^*)$ for any $\mathbf{x} \in C \cap B[\mathbf{x}^*, r]$. Let $\mathbf{x}^* \neq \mathbf{y} \in C$. We will show that $f(\mathbf{y}) \geq f(\mathbf{x}^*)$. Let $\lambda \in (0, 1)$ be such that $\mathbf{x}^* + \lambda(\mathbf{y} - \mathbf{x}^*) \in B[\mathbf{x}^*, r]$. Since $\mathbf{x}^* + \lambda(\mathbf{y} - \mathbf{x}^*) \in B[\mathbf{x}^*, r]$, it follows that $f(\mathbf{x}^*) \leq f(\mathbf{x}^* + \lambda(\mathbf{y} - \mathbf{x}^*))$ and hence by Jensen's inequality:

$$f(\mathbf{x}^*) \leq f(\mathbf{x}^* + \lambda(\mathbf{y} - \mathbf{x}^*)) \leq (1 - \lambda)f(\mathbf{x}^*) + \lambda f(\mathbf{y}).$$

Thus, the desired inequality $f(\mathbf{x}^*) \leq f(\mathbf{y})$ follows. □

A small variation of the proof of the last theorem yields the following.

Theorem. *Let $f : C \to \mathbb{R}$ be a strictly convex function defined on the convex set C. Let $x^* \in C$ be a local minimum of f over C. Then x^* is a strict global minimum of f over C.*

Another important and easily deduced property of convex problems is that set of optimal solutions is also convex.

Theorem. *Let $f : C \to \mathbb{R}$ be a convex function defined over the convex set $C \subseteq \mathbb{R}^n$. Then the set of optimal solutions of the problem*

$$\min\{f(\mathbf{x}) : \mathbf{x} \in C\}$$

is convex. If, in addition, f is strictly convex over C, then there exists at most one optimal solution of the problem.

Examples:

- A Convex Problem:

$$\begin{array}{ll} \min & -2x_1 + x_2 \\ \text{s.t.} & x_1^2 + x_2^2 \leq 3 \end{array}$$

- A Nonconvex Problem:

$$\begin{array}{ll} \min & x_1^2 - x_2 \\ \text{s.t.} & x_1^2 + x_2^2 = 3 \end{array}$$

- Linear Programming

$$(\mathbf{LP}): \quad \begin{array}{ll} \min & \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} & \mathbf{A}\mathbf{x} \leq \mathbf{b} \\ & \mathbf{B}\mathbf{x} = \mathbf{g} \end{array}$$

  (LP) is a convex optimization problem (constraints and objective function are linear / affine and hence convex). It is also equivalent to a problem of maximizing a convex (linear) function subject to a convex constraints set. Hence, if the feasible set is compact and nonempty, then there exists at least one optimal solution which is an extreme point, or equivalently, a basic feasible solution.

- Convex Quadratic Problems consist of minimizing a convex quadratic function subject to affine constraints. The general form is

$$\begin{array}{ll} \min & \mathbf{x}^\top \mathbf{Q}\mathbf{x} + 2\mathbf{b}^\top \mathbf{x} \\ \text{s.t.} & \mathbf{A}\mathbf{x} \le \mathbf{c} \end{array}$$

  $\mathbf{Q} \in \mathbb{R}^{n\times n}$ is positive semidefinite, $\mathbf{b} \in \mathbb{R}^n$, $\mathbf{A} \in \mathbb{R}^{m\times n}$, $\mathbf{c} \in \mathbb{R}^m$.

Optimization over a Convex Set and Stationarity

We will consider the constrained optimization problem given by

$$\min_{\mathbf{x}} \quad \{f(\mathbf{x}) : \mathbf{x} \in C\}, \tag{P}$$

where C is a closed convex subset of $\mathbb{R}^n$, and f is continuously differentiable over C, not necessarily convex. To characterize optimality in the presence of convex constraints, we define the concept of stationarity.

Definition (Stationarity)**.** *Let f be a continuously differentiable function over a closed and convex set C. Then $\mathbf{x}^*$ is called a stationary point of* (P) *if*

$$\nabla f(\mathbf{x}^*)^\top (\mathbf{x} - \mathbf{x}^*) \ge 0, \qquad \textit{for any } \mathbf{x} \in C.$$

Theorem (Stationarity as a Necessary Optimality Condition)**.** *Let f be a continuously differentiable function over a nonempty closed convex set C, and let $\mathbf{x}^*$ be a local minimum of* (P)*. Then $\mathbf{x}^*$ is a stationary point of* (P).

Proof. Let $\mathbf{x}^*$ be a local minimum of (P), and assume in contradiction that $\mathbf{x}^*$ is not a stationary point of (P). This implies that there exists $\mathbf{x} \in C$ such that

$$\nabla f(\mathbf{x}^*)^\top (\mathbf{x} - \mathbf{x}^*) < 0.$$

Thus, $f'(\mathbf{x}^*; \mathbf{d}) < 0$, where $\mathbf{d} = \mathbf{x} - \mathbf{x}^*$. Therefore, there exists $\varepsilon \in (0,1)$ such that $f(\mathbf{x}^* + t\mathbf{d}) < f(\mathbf{x}^*)$, $\forall t \in (0,\varepsilon)$. Finally, since $\mathbf{x}^* + t\mathbf{d} = (1-t)\mathbf{x}^* + t\mathbf{x} \in C$, $\forall t \in (0,\varepsilon)$, we conclude that $\mathbf{x}^*$ is not a local optimum point of (**P**). Contradiction. □

Examples of Stationarity Conditions

- For $C = \mathbb{R}^n$, $\mathbf{x}^*$ is a stationary point of (P) iff

$$\nabla f(\mathbf{x}^*)^\top (\mathbf{x} - \mathbf{x}^*) \ge 0 \quad \forall \mathbf{x} \in \mathbb{R}^n.$$

  We will show that the above condition is equivalent to $\nabla f(\mathbf{x}^*) = 0$, Indeed, if $\nabla f(\mathbf{x}^*) = 0$, then obviously the inequality is satisfied. In the other direction, suppose that the inequality holds. Plugging $\mathbf{x} = \mathbf{x}^* - \nabla f(\mathbf{x}^*)$ in the above implies

$$-\|\nabla f(\mathbf{x}^*)\|^2 \ge 0.$$

  Thus, $\nabla f(\mathbf{x}^*) = 0$.

- For $C = \mathbb{R}^n_+$, $\mathbf{x}^*$ is a stationary point iif

  $$\nabla f\left(\mathbf{x}^*\right)^\top \left(\mathbf{x} - \mathbf{x}^*\right) \geq 0 \quad \forall \mathbf{x} \in \mathbb{R}^n_+ .$$

  This is equivalent to say $\nabla f\left(\mathbf{x}^*\right)^\top \mathbf{x} - \nabla f\left(\mathbf{x}^*\right)^\top \mathbf{x}^* \geq 0$ for all $\mathbf{x} \geq 0$. Or equivalently, $\nabla f\left(\mathbf{x}^*\right) \geq 0$ and $\nabla f\left(\mathbf{x}^*\right)^\top \mathbf{x}^* \leq 0$. The latter is equivalent to claim that

  $$\nabla f\left(\mathbf{x}^*\right) \geq 0 \qquad \text{and} \qquad x_i^* \frac{\partial f}{\partial x_i}\left(\mathbf{x}^*\right) = 0, \qquad i = 1, 2, \ldots, n,$$

  which we summarize as

  $$\frac{\partial f}{\partial x_i}\left(\mathbf{x}^*\right) \begin{cases} = 0 & x_i^* > 0 \\ \geq 0 & x_i^* = 0 \end{cases}$$

Some important stationarity conditions are summarized in the table below:

Feasible Set	Explicit Stationarity Condition
$\mathbb{R}^n$	$\nabla f\left(\mathbf{x}^*\right) = \mathbf{0}$
$\mathbb{R}^n_+$	$\frac{\partial f}{\partial x_i}\left(\mathbf{x}^*\right) \begin{cases} = 0 & x_i^* > 0 \\ \geq 0 & x_i^* = 0 \end{cases}$
$\{\mathbf{x} \in \mathbb{R}^n : \mathbf{e}^\top \mathbf{x} = 1\}$	$\frac{\partial f}{\partial x_1}\left(\mathbf{x}^*\right) = \ldots = \frac{\partial f}{\partial x_n}\left(\mathbf{x}^*\right)$
$B[\mathbf{0}, 1]$	$\nabla f\left(\mathbf{x}^*\right) = \mathbf{0}$ or $\lVert\mathbf{x}^*\rVert = 1$ and $\exists \lambda \leq 0 : \nabla f\left(\mathbf{x}^*\right) = \lambda \mathbf{x}^*$

For convex problems, stationarity is a necessary and sufficient condition.

Theorem (Stationarity in Convex Optimization). *Let f be a continuously differentiable convex function over a nonempty closed and convex set $C \subseteq \mathbb{R}^n$. Then $\mathbf{x}^*$ is a stationary point of* (P) *iff $\mathbf{x}^*$ is an optimal solution of* (P)*.*

Proof. If $\mathbf{x}^*$ is an optimal solution of (P), then we already showed that it is a stationary point of (P). Assume that $\mathbf{x}^*$ is a stationary point of (P). Let $\mathbf{x} \in C$. Then

$$f(\mathbf{x}) \geq f\left(\mathbf{x}^*\right) + \nabla f\left(\mathbf{x}^*\right)^\top \left(\mathbf{x} - \mathbf{x}^*\right) \geq f\left(\mathbf{x}^*\right)$$

establishing the optimality of $\mathbf{x}^*$. □

The Orthogonal Projection Operator

Definition (Orthogonal Projection). *Given a nonempty closed convex set C, the orthogonal projection operator $P_C : \mathbb{R}^n \to C$ is defined by*

$$P_C(\mathbf{x}) = \operatorname{argmin}\left\{\lVert\mathbf{y} - \mathbf{x}\rVert^2 : \mathbf{y} \in C\right\} .$$

We state two important results concerning orthogonal projections.

Theorem (The First Projection Theorem). *Let $C \subseteq \mathbb{R}^n$ be a nonempty closed and convex set. Then for any $\mathbf{x} \in \mathbb{R}^n$, the orthogonal projection $P_C(\mathbf{x})$ exists and is unique.*

Theorem (The Second Projection Theorem). *Let C be a nonempty closed convex set and let $\mathbf{x} \in \mathbb{R}^n$. Then $\mathbf{z} = P_C(\mathbf{x})$ if and only if*

$$(\mathbf{x} - \mathbf{z})^\top (\mathbf{y} - \mathbf{z}) \leq 0\,, \qquad \textit{for any } \mathbf{y} \in C$$

Examples of Orthogonal Projections:

- For $C = \mathbb{R}^n_+$,

$$P_{\mathbb{R}^n_+}(\mathbf{x}) = [\mathbf{x}]_+$$

  where $[\mathbf{v}]_+ = (\max\{v_1, 0\}, \max\{v_2, 0\}, \ldots, \max\{v_n, 0\})^\top$.

- A box is a subset of $\mathbb{R}^n$ of the form

$$B = [\ell_1, u_1] \times [\ell_2, u_2] \times \cdots \times [\ell_n, u_n] = \{\mathbf{x} \in \mathbb{R}^n : \ell_i \leq x_i \leq u_i\}\,,$$

  where $\ell_i \leq u_i$ for all $i = 1, 2, \ldots, n$. For this set

$$[P_B(\mathbf{x})]_i = \begin{cases} u_i & x_i \geq u_i \\ x_i & \ell_i < x_i < u_i \\ \ell_i & x_i \leq \ell_i \end{cases}$$

- For the closed ball in $\mathbb{R}^n$, $C = B[0, r]$, it holds

$$P_{B[0,r]} = \begin{cases} \mathbf{x} & \|\mathbf{x}\| \leq r \\ r\frac{\mathbf{x}}{\|\mathbf{x}\|} & \|\mathbf{x}\| > r \end{cases}$$

A very important result in convex optimization is the representation of stationarity using the orthogonal projection operator.

Theorem (Representation of Stationarity via the Orthogonal Projection Operator). *Let f be a continuously differentiable function over the nonempty closed convex set C, and let $s > 0$. Then $\mathbf{x}^*$ is a stationary point of* (P) *if and only if*

$$\mathbf{x}^* = P_C\left(\mathbf{x}^* - s\nabla f\left(\mathbf{x}^*\right)\right)\,.$$

Proof. By the second projection theorem, $\mathbf{x}^* = P_C\left(\mathbf{x}^* - s\nabla f\left(\mathbf{x}^*\right)\right)$ iff

$$\left(\mathbf{x}^* - s\nabla f\left(\mathbf{x}^*\right) - \mathbf{x}^*\right)^\top \left(\mathbf{x} - \mathbf{x}^*\right) \leq 0 \text{ for any } \mathbf{x} \in C\,,$$

which is equivalent to

$$\nabla f\left(\mathbf{x}^*\right)^\top \left(\mathbf{x} - \mathbf{x}^*\right) \geq 0 \text{ for any } \mathbf{x} \in C\,,$$

namely, the definition of stationarity. □

The Gradient Projection Method

It is convenient to define the gradient mapping as

$$G_L(\mathbf{x}) = L\left[\mathbf{x} - P_C\left(\mathbf{x} - \frac{1}{L}\nabla f(\mathbf{x})\right)\right],$$

where $L > 0$. In the unconstrained case $G_L(\mathbf{x}) = \nabla f(\mathbf{x})$. Otherwise, $G_L(\mathbf{x}) = \mathbf{0}$ if and only if $\mathbf{x}$ is a stationary point of (P). This means that we can consider $\|G_L(\mathbf{x})\|^2$ to be optimality measure. We use the orthogonal projection operator to introduce a gradient descent type algorithm for solving convex optimization problems.

Algorithm 1: The Gradient Projection Method

Initialization: A tolerance parameter $\varepsilon > 0$ and $\mathbf{x}^0 \in C$.
General Step: for any $k = 0, 1, 2, \ldots$ execute the following steps:
Pick a stepsize t^k by a line search procedure.
Set $\mathbf{x}^{k+1} = P_C\left(\mathbf{x}^k - t^k\nabla f\left(\mathbf{x}^k\right)\right)$.
If $\left\|\mathbf{x}^k - \mathbf{x}^{k+1}\right\| \le \varepsilon$, then STOP and $\mathbf{x}^{k+1}$ is the output.

There are several strategies for choosing the stepsizes t^k. When $f \in C_L^{1,1}$, we can choose t^k to be constant and equal to $\frac{1}{L}$. An alternative is to include backtracking.

Algorithm 2: The Gradient Projection Method with Backtracking

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Initialization: A tolerance parameter $\varepsilon > 0$ and $\mathbf{x}^0 \in C$. Parameters $s > 0$, $\alpha \in (0, 1)$, and $\beta \in (0, 1)$.
General Step: for any $k = 0, 1, 2, \ldots$ execute the following steps:
Pick $t^k = s$.
While $f(\mathbf{x}^k) - f(P_C(\mathbf{x}^k - t^k\nabla f(\mathbf{x}^k))) < \alpha t^k \left\|G_{\frac{1}{t^k}}(\mathbf{x}^k)\right\|^2$, set $t^k := \beta t^k$.
Set $\mathbf{x}^{k+1} = P_C\left(\mathbf{x}^k - t^k\nabla f\left(\mathbf{x}^k\right)\right)$.
If $\left\|\mathbf{x}^k - \mathbf{x}^{k+1}\right\| \le \varepsilon$, then STOP and $\mathbf{x}^{k+1}$ is the output.

Theorem (Convergence of the Gradient Projection Method)**.** *Let $\left\{\mathbf{x}^k\right\}$ be the sequence generated by the gradient projection method for solving problem* (P) *with either a constant stepsize $\bar{t} \in \left(0, \frac{2}{L}\right)$, where L is a Lipschitz constant of ∇f or a backtracking stepsize strategy. Assume that f is bounded below. Then:*

1. *The sequence $\left\{f\left(\mathbf{x}^k\right)\right\}$ is nonincreasing.*
2. *$G_d\left(\mathbf{x}^k\right) \to 0$ as $k \to \infty$, where*

$$d = \begin{cases} 1/\bar{t} & \textit{constant stepsize} \\ 1/s & \textit{backtracking.} \end{cases}$$

The Frank-Wolfe Method

The Frank-Wolfe method, also known as the conditional gradient method, was developed as an alternative to the projected gradient method when the projection operator is not explicitly defined.

Let us revisit the following problem:

$$\min \{f(\mathbf{x}) : \mathbf{x} \in C\}$$

where $f\colon \mathbb{R}^n \to \mathbb{R}$ is a continuously differentiable function, and C is a convex and compact set.

In many cases, computing the projection operator requires solving an additional problem, which can be as computationally intensive as the original problem. In contrast, the Frank-Wolfe method leverages the fact that minimising a linear objective over a convex and compact set is relatively straightforward. To illustrate this point, let us consider an example:

Example: Let $C = \{\mathbf{x} \in \mathbb{R}^n \colon A\mathbf{x} \leq \mathbf{b}\}$ be the convex polytope defined by $A \in \mathbb{R}^{m\times n}$ and $\mathbf{b} \in \mathbb{R}^m$. The orthogonal projection over C is defined as the problem

$$P_C(\mathbf{x}) = \underset{\mathbf{y}\in\mathbb{R}^n}{\operatorname{argmin}} \left\{\|\mathbf{y}-\mathbf{x}\|^2 \colon A\mathbf{y} \leq \mathbf{b}\right\}$$

Unfortunately, there is no explicit solution for this problem. To solve it, one can use the KKT conditions (Chapter 7) or duality theory (Chapter 8) to obtain a parameter-dependent solution.

In contrast, the following problem:

$$\min_{\mathbf{y}\in\mathbb{R}^n} \left\{\mathbf{a}^\intercal\mathbf{y} \colon\ A\mathbf{y} \leq \mathbf{b}\right\}$$

with $\mathbf{a} \in \mathbb{R}^n$, is a *linear programming problem*. It can be shown that the solution is one of the extreme points (vertices) of the polytope. This problem can be solved efficiently using the Simplex Method.

Next, we describe the Frank-Wolfe method for solving problem P. The core idea is to iteratively approximate the objective function $f(\mathbf{x})$ by its first-order Taylor expansion around the current iterate and solve the following subproblem:

$$\begin{aligned}\bar{\mathbf{x}}^k &:= \underset{\bar{\mathbf{x}}\in C}{\operatorname{argmin}}\, f(\mathbf{x}^k) + \nabla f(\mathbf{x}^k)^\intercal(\bar{\mathbf{x}} - \mathbf{x}^k) \\ &= \underset{\bar{\mathbf{x}}\in C}{\operatorname{argmin}}\, \nabla f(\mathbf{x}^k)^\intercal\bar{\mathbf{x}}.\end{aligned} \tag{(FW-P)}$$

The next iteration is then updated as:

$$\mathbf{x}^{k+1} = \mathbf{x}^k + \alpha_k(\bar{\mathbf{x}}^k - \mathbf{x}^k),$$

with $\alpha_k \in [0,1]$.

Remark

- The method is typically applied to problems where C is compact, ensuring the subproblem (FW-P) is well-posed due to Weierstrass' Theorem (Chapter 2).
- The feasibility of the iterates is guaranteed, provided the initial guess is feasible, i.e., $\mathbf{x}^0 \in C$ thanks to the convexity of C.
- In the original version of the method (1956), they proposed the following stepsize rule: $\alpha_k = 2/k{+}2$, for $k = 1, 2, \dots$.

Algorithm 3: The Frank-Wolfe Method

Initialization: $\mathbf{x}^0 \in C$.

General Step: for any $k = 0, 1, 2, \dots$ execute the following steps:

1 Find $\bar{\mathbf{x}}^k$ by solving problem (FW-P)

2 Set $\alpha_k = 2/k{+}2$

3 Update $\mathbf{x}^{k+1} = \mathbf{x}^k + \alpha_k(\bar{\mathbf{x}}^k - \mathbf{x}^k)$

Theorem (Convergence of the Frank-Wolfe Method)**.** *Suppose that f is a convex function whose gradient is Lipschitz continuously differentiable with constant L on an open neighbourhood of C, where C is a closed, bounded and convex set with diameter $D := \max_{\mathbf{x},\mathbf{y}\in C}\|\mathbf{y}-\mathbf{x}\|$, and let $\mathbf{x}^*$ be the solution to problem P. Moreover, let $\{\mathbf{x}^k\}$ be the sequence generated by the Frank-Wolfe method, then we have*

$$f(\mathbf{x}^k) - f(\mathbf{x}^*) \leq \frac{2LD^2}{k+2} \qquad k = 1, 2, \dots .$$

We end this chapter with an illustration of the difference between the gradient projection method and the Frank-Wolfe method.

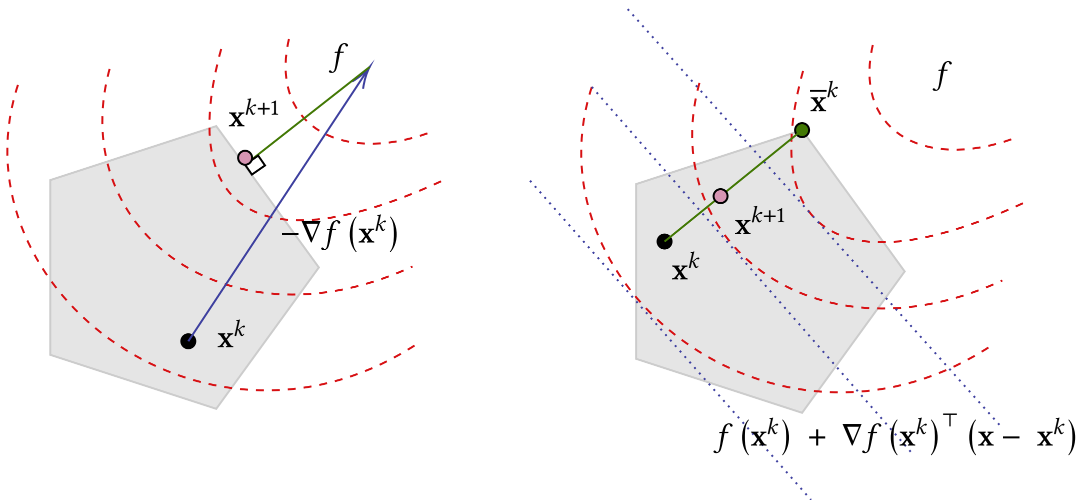

Figure 1: The gradient projection method (left) uses an orthogonal projection to guarantee feasibility and the Frank-Wolfe method (right), ensures feasibility by using convex combinations.